

JETSETGO



CCSW 315 Software Process Models

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Project Initiation

The application is a comprehensive airport service platform designed to enhance the overall travel experience for passengers by addressing various challenges encountered during air travel. A key feature is the personal companion service, which is not only for unaccompanied minors but also for individuals with special needs and the elderly who may require additional support while traveling alone. This service ensures safety and provides dedicated assistance, giving families peace of mind. Users can also explore and reserve airport lounges in advance, gaining access to comfortable environments with various amenities. Furthermore, it addresses parking challenges by allowing users to reserve parking spaces before arrival, thus reducing stress and uncertainty. Lastly, the application offers a private driver service, facilitating convenient and reliable transportation options for travelers upon their arrival. By integrating these features, the app aims to streamline airport operations, improve communication, and ultimately create a more enjoyable and efficient travel experience for all passengers.

SCOP

The primary objective of this application is to streamline airport operations and improve passenger experience by providing timely information, personalized services, and convenient solutions. By addressing the common pain points associated with air travel, the app aims to make the journey more enjoyable and less stressful for all users.

Key Features of the Application:

01. Personal Companion Service:

A dedicated service for unaccompanied minors, individuals with special needs, and the elderly ensures that all travelers receive personalized support and guidance, addressing safety concerns for families and caregivers

02. Lounge Reservations:

Users can discover available lounges, access amenities, and reserve spaces in advance, providing a comfortable environment to relax before flights

03. Parking Solutions:

The app simplifies the parking process by allowing users to reserve spots ahead of time, reducing stress and uncertainty upon arrival at the airport

04. Private Driver Services:

Travelers can book private transportation through the app, ensuring a smooth and hassle-free journey to or from the airport.

Users:

1. TRAVELING FAMILIES:

Parents traveling with children, particularly those with unaccompanied minors, will benefit from the personal companion service, ensuring safety and support during the journey

2. INDIVIDUALS WITH SPECIAL NEEDS:

Travelers who require assistance due to physical or cognitive challenges will find the companion service invaluable, providing tailored support and guidance

3. FREQUENT FLYERS:

Business travelers and frequent flyers will appreciate streamlined services and convenient access to lounges, enhancing their overall travel experience

4. LEISURE TRAVELERS:

Tourists and casual travelers looking for a seamless airport experience will enjoy the convenience of parking reservations, and private transportation options.

5. AIRPORT STAFF AND AUTHORITIES:

Airport personnel may also utilize the application for operational insights and to improve service delivery

Problems the APP Will Solve:

1. Safety and Support for Minors:

- Problem: Families are often concerned about the safety of children, individuals with special needs, and the elderly traveling alone.
- Solution: A personal companion service provides dedicated support and guidance for unaccompanied minors, individuals with special needs, and the elderly, offering peace of mind to families and caregivers.

2. Lounge Access and Information:

- Problem: Travelers may not know about lounge options or how to access them.
- Solution: The app allows users to reserve lounges and provides detailed information about amenities and services, enhancing comfort during layovers

3. Parking Challenges:

- Problem: Finding parking at busy airports can be time-consuming and stressful.
- Solution: The app enables users to reserve parking spaces in advance, ensuring a hasslefree arrival.

4. Transportation Options:

- Problem: Travelers often face difficulties in arranging reliable transportation upon arrival.
- Solution: A private driver service provides convenient and dependable transport options for passengers.

Prioritized Functional requirements:

Component Name	Login
Story Name	Login
Story Sequence No	001
Story Short Description	Users must be able to log into the application using their email and password, with an option to reset the password.
Story Long Description	The system shall allow users to enter their credentials (email and password) to access the application.

Component Name	Sign Up
Story Name	Sign Up
Story Sequence No	002
Story Short Description	Users must be able to create a new account by providing their personal information.
Story Long Description	The system shall allow users to register for an account by entering the following information: Name, Age, Email, Phone Number, Password. Upon submitting the registration form: If the email is already associated with an existing account, an error message will inform the user: 'failed to sign up. Please try again.' If all information is valid, the system will save the user's information in the database and redirect them to the main page.

Component Name	Private Driver Services
Story Name	Private Driver Services
Story Sequence No	003
Story Short Description	Arrange private transportation.
Story Long Description	The system shall allow travelers to book private transportation services from or to the airport.

Component Name	Parking Reservations
Story Name	Parking Reservation
Story Sequence No	004
Story Short Description	Users can reserve airport parking.
Story Long Description	The system shall allow users to select and reserve parking spaces in advance. Users will be able to provide details such as name, phone number, email address, vehicle number, arrival date and time, reservation period, and select a parking spot from a visual grid. All reservations will be stored in the database.

Component Name	Lounge Reservations
Story Name	Lounge Reservation
Story Sequence No	005
Story Short Description	Users can search for and reserve airport lounges.
Story Long Description	The system shall allow users to search for available airport lounges, input their personal information, select a desired lounge and departure time, and complete the booking process. The booking confirmation shall be displayed to the user, and the reservation details shall be stored in the user's profile for easy access.

Component Name	Personal Companion Service
Story Name	Personal Companion Service
Story Sequence No	006
Story Short Description	Offer support services for travelers with special needs.
Story Long Description	The service shall assist elderly travelers, unaccompanied minors, and passengers with special needs to ensure their safety and support while navigating the airport. The application shall allow users to request this service by providing their details, such as name, email, phone number, and arrival date/time. Users will also be able to select the type of traveler (e.g., child or adult) through a toggle button. Once the request is submitted, airport staff will be notified to provide the necessary assistance.

Prioritized Non-Functional requirements:

Component Name	Security
Story Name	Secure User Data
Story Sequence No	N-FR001
Story Short Description	The application must ensure the security of user data.
Story Long Description	The system shall be designed with robust security measures to protect user data, including personal, and travel-related information, from unauthorized access. Encryption and secure authentication mechanisms shall be implemented to safeguard sensitive data.

Component Name	User Interface
Story Name	Simple and Intuitive User Interface
Story Sequence No	N-FR002
Story Short Description	The application should have a user-friendly interface.
Story Long Description	The system shall provide a simple, intuitive, and easy-to-use user interface that accommodates users of all age groups. Users should be able to easily navigate the app's features without prior training or extensive technical knowledge.

Component Name	Responsiveness
Story Name	Real-Time Responsiveness
Story Sequence No	N-FR003
Story Short Description	The application must provide real-time updates.
Story Long Description	The system shall deliver real-time responses without delay. The responsiveness of the system shall be optimized to ensure minimal lag and accurate data delivery.

Component Name	Availability
Story Name	24/7 Availability
Story Sequence No	N-FR004
Story Short Description	The application must be available at all times.
Story Long Description	The system shall be operational 24/7 to ensure travelers can access critical information and services at any time, whether day or night, without interruptions or downtime.

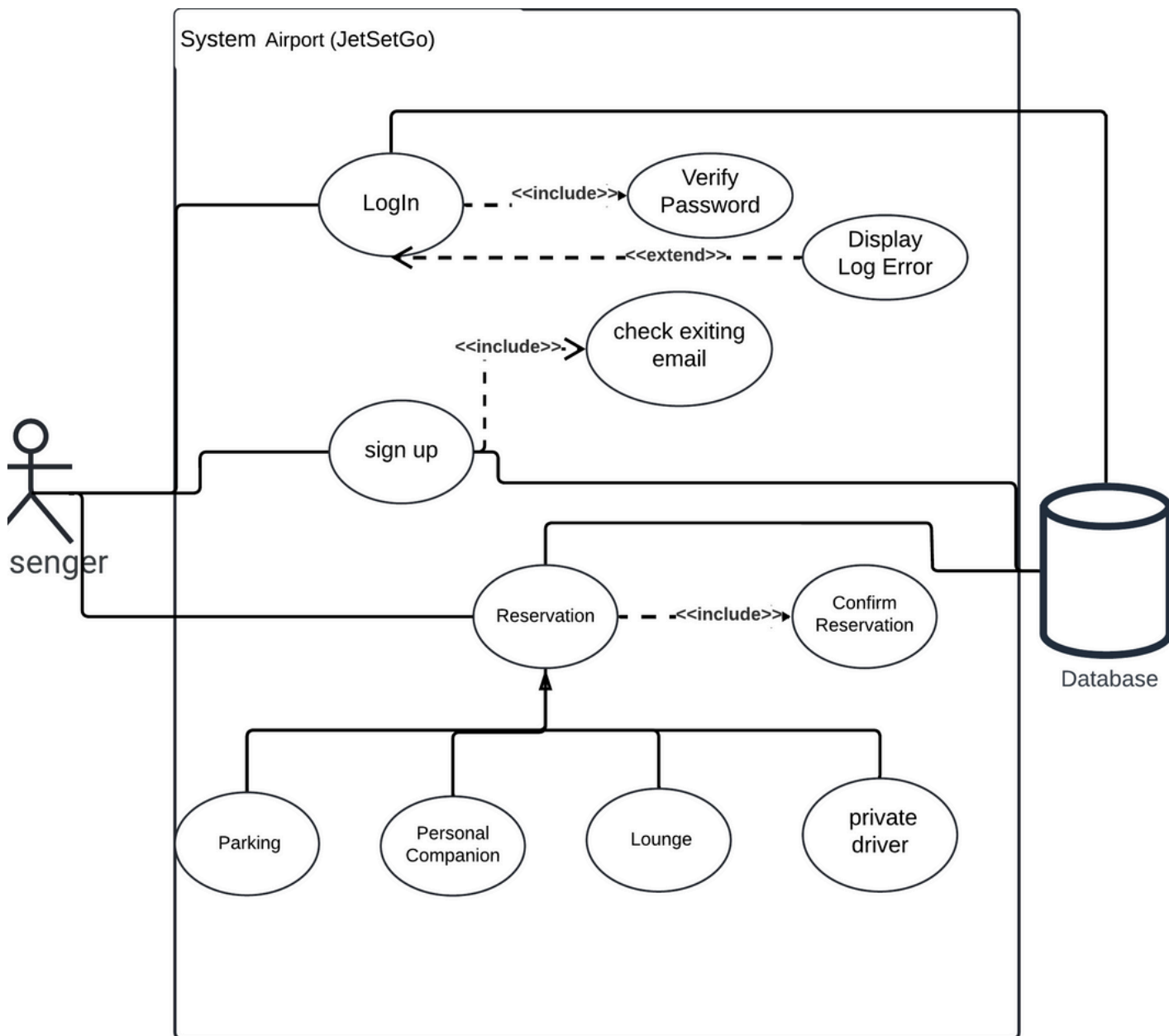
Component Name	Scalability
Story Name	Scalable Design
Story Sequence No	N-FR005
Story Short Description	The application must support scalability.
Story Long Description	The system shall be designed to scale effectively, accommodating an increasing number of users and airport data without degrading performance. The architecture shall support the growth of the application without causing system slowdowns or failures.

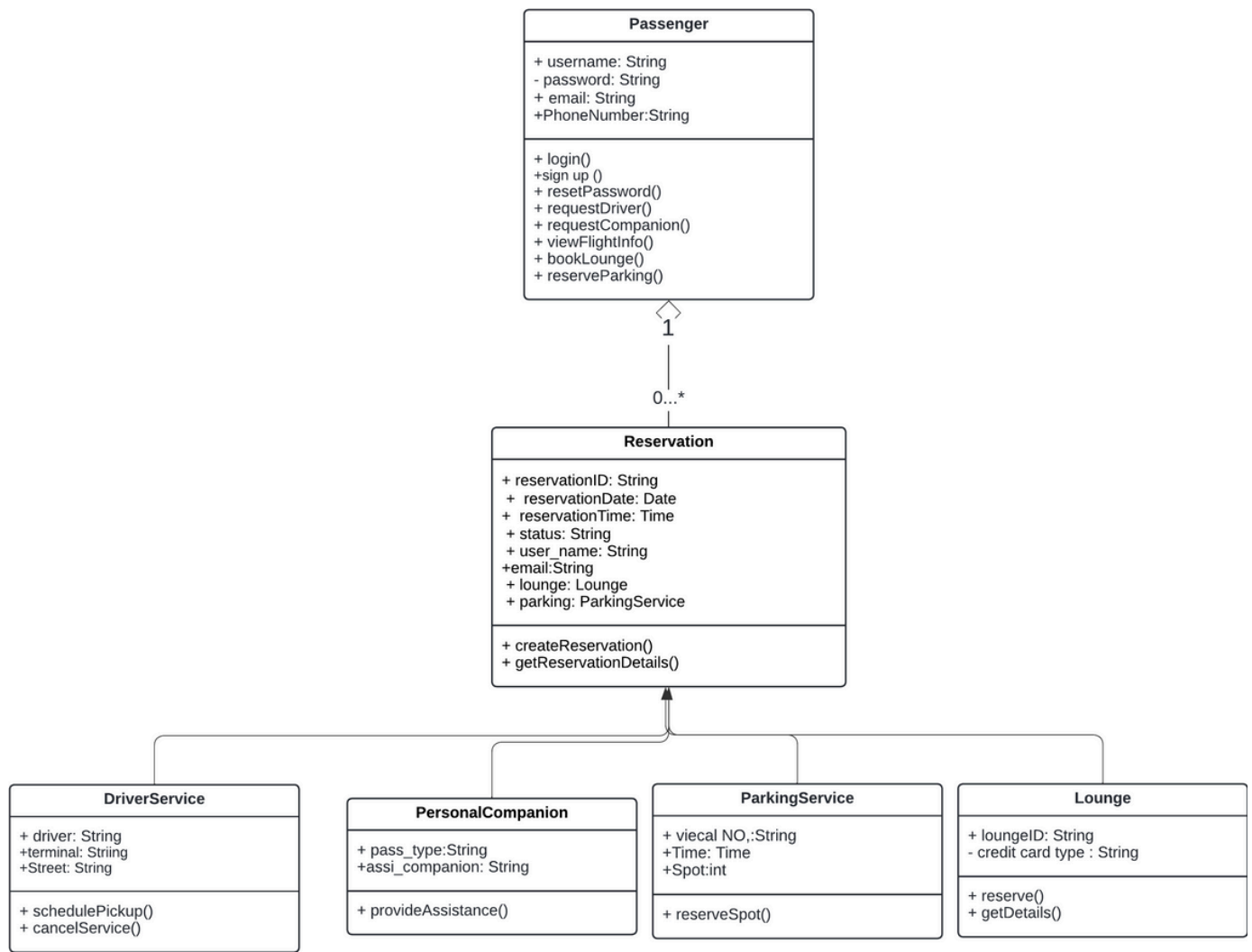
Component Name	Reliability
Story Name	Reliable System Performance
Story Sequence No	N-FR006
Story Short Description	The application must provide reliable data.
Story Long Description	The system shall be reliable, consistently delivering accurate and timely data without errors, interruptions, or delays. Users must be able to depend on the app for critical travel-related information.

Component Name	Compatibility
Story Name	Multi-Platform Compatibility
Story Sequence No	N-FR007
Story Short Description	The application must work across multiple platforms.
Story Long Description	The system shall be compatible with various operating systems, including iOS and Android, and perform efficiently across different mobile devices. The app shall be optimized for a wide range of screen sizes and hardware configurations.

Component Name	Continuous Updates and Development
Story Name	Continuous Development
Story Sequence No	N-FR008
Story Short Description	The application must undergo regular updates.
Story Long Description	The system shall be continuously updated to improve existing services, enhance performance, and introduce new features. Regular updates will ensure users experience consistent improvements.

Use Case and Class Diagram:





Tools:

IDE: Visual Studio (dart/flutter)

Agile Planning: JIRA

Modeling: Lucidchar

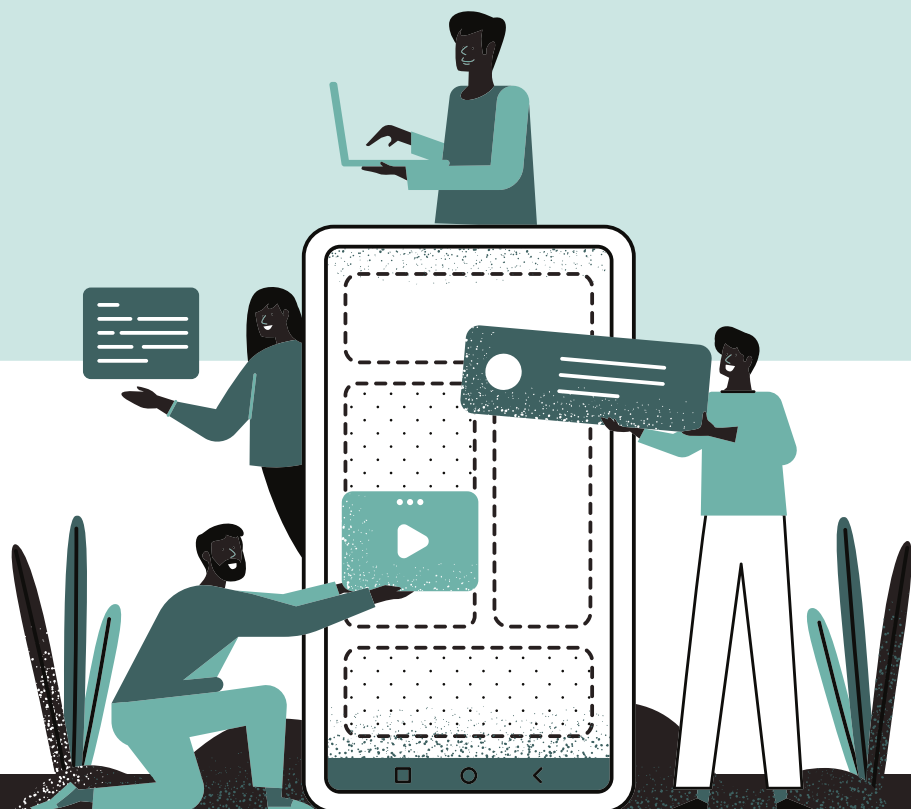
Git: GitHub

Messaging: Slack

backend: Node.js

Databased: MySQL

SPRINT ONE



1. What has been completed since the last meeting?

1. Sign Up Feature:

- Developed the user sign-up functionality where users can create a new account by providing their personal details (name, age, email, phone number, and password).
- Implemented backend validation for sign-up input fields and integrated password hashing using bcrypt for secure storage.

2. Driver Reservation Feature:

- Implemented the backend API to allow users to make driver reservations. Users can provide details such as the pickup city, street, postcode, terminal, and pickup time.
- Created the driver reservation UI in Flutter, allowing users to enter reservation details and submit them to the backend.

2. What are you going to be working on next?

- Log In Feature: Start developing the login functionality, allowing users to authenticate and access their account using their email and password.
- Lounge Reservation Feature: Begin working on the lounge reservation feature, allowing users to browse and book airport lounges.

3. Do you have any issues/impediments?

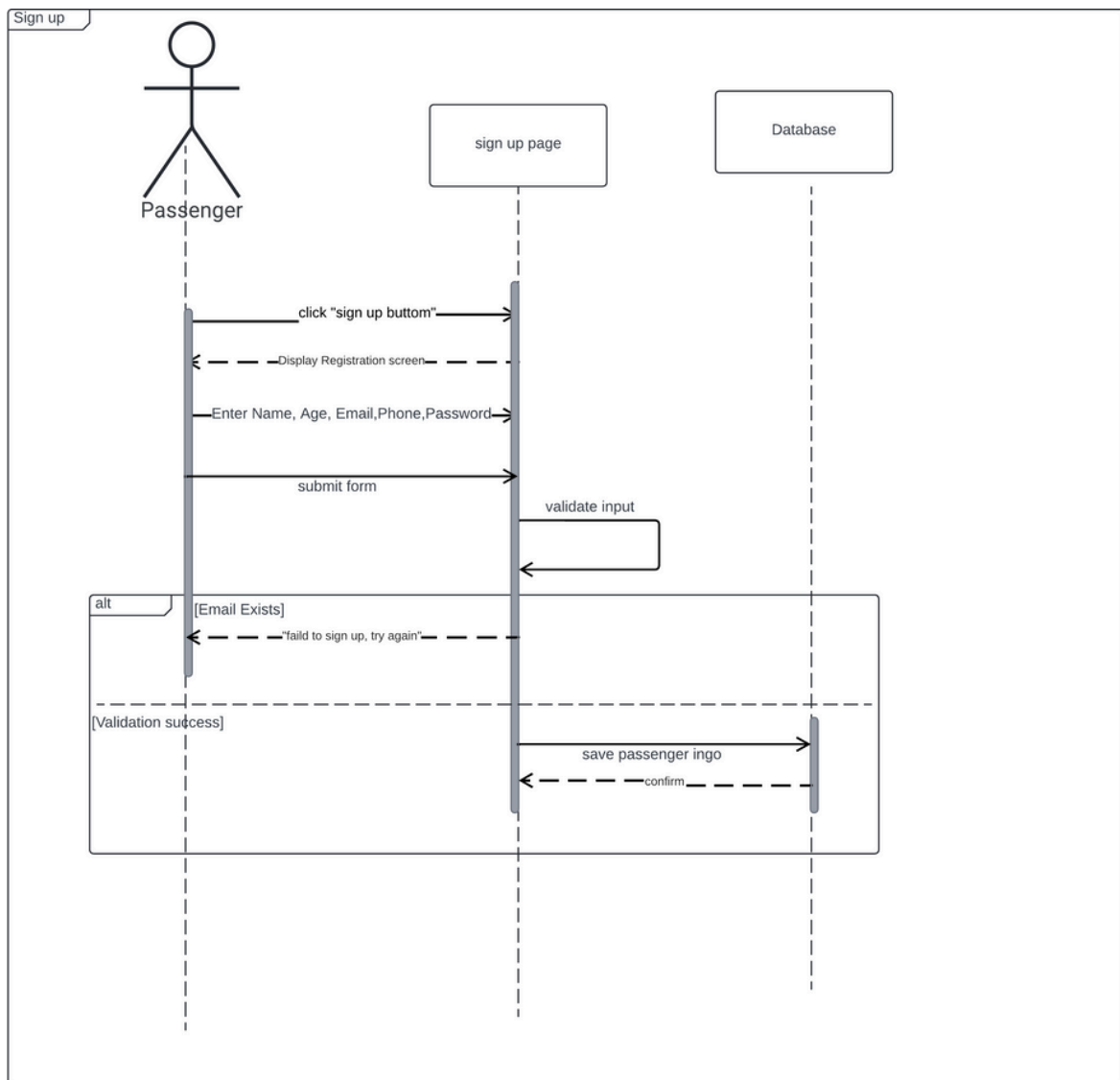
- Session Management: Ensuring user sessions are properly maintained across screens after sign-up can be tricky, especially with managing the email and session state in Flutter.
- Backend Validation: Some validation errors occurred during the sign-up process that need further attention for consistency and better user feedback.

Story backlog:

Story Name	Sign Up
Story Sequence No	002
Story Short Description	Users must be able to create a new account by providing their personal information.
Story Long Description	The system shall allow users to register for an account by entering the following information: Name, Age, Email, Phone Number, Password. Upon submitting the registration form: If the email is already associated with an existing account, an error message will inform the user: "Failed to sign up. Please try again." If all information is valid, the system will save the user's information in the database and redirect them to the main page.

Test Case Name: [Sprint 1 – Sign Up]

Test Case NO.	Test Case Description	Expected Results	Outcome (Pass/Fail/Other(Comments))
TC001	Passenger selects "Sign Up"	Display registration screen	
TC002	Passenger enters information Name, Age, Email, Phone Number, Password	Depending on missing fields or valid input, different outcomes occur	
TC003	User enters existing email address	User should be displayed with a message: "Failed to Sign Up. Please try again."	
TC004	User enters valid information (Name, Age, Email, Phone Number, Password)	Save information in the database and show main page	

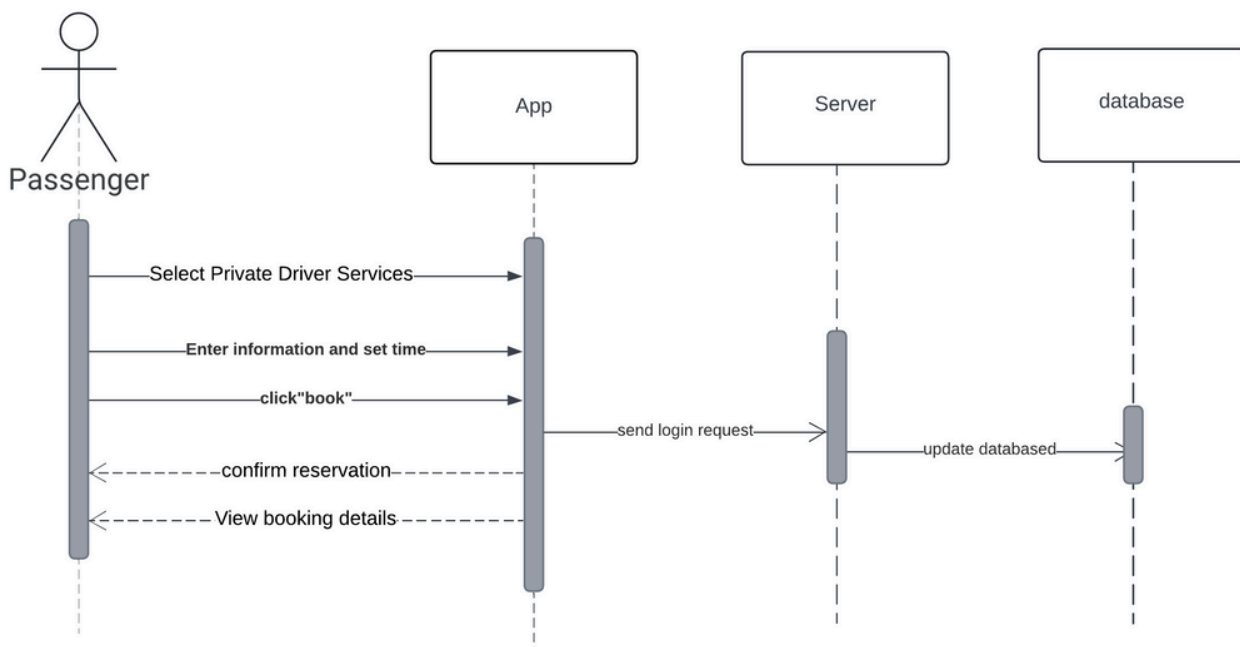


Story Name	Private Driver Services
Story Sequence No	003
Story Short Description	Arrange private transportation.
Story Long Description	The system shall allow travelers to book private transportation services from or to the airport.

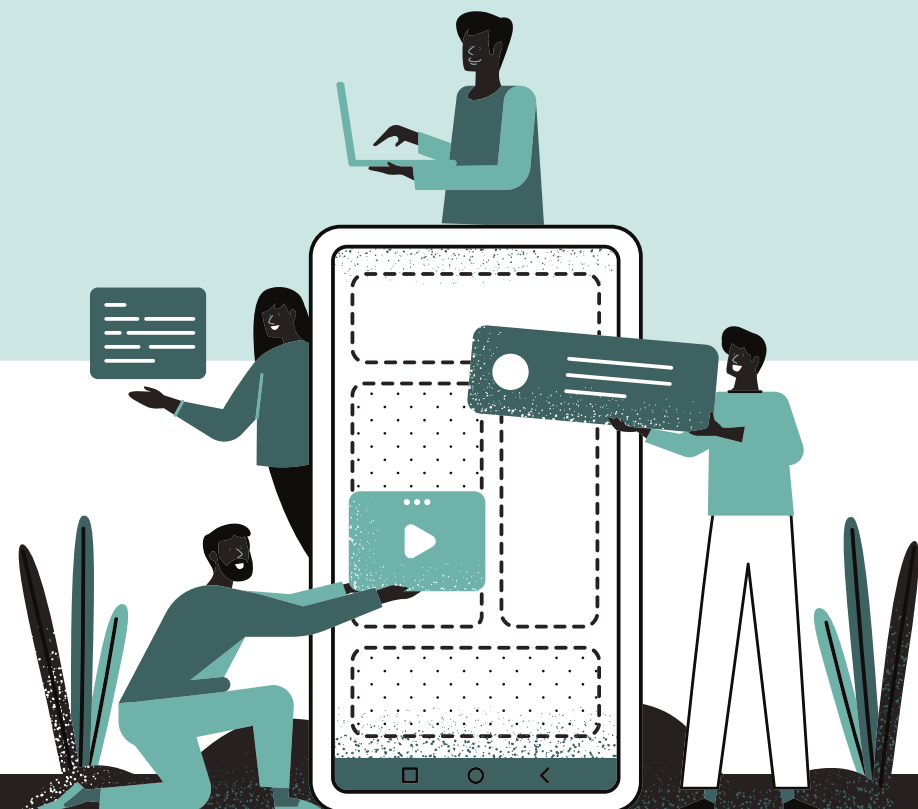
Test Case Name: [Sprint 1 – driver Services]

Test Case NO.	Test Case Description	Expected Results	Outcome (Pass/ Fail/ Other(Comments))
TC001	User selects "Private Driver Services" in the app	The system should display the page that allows the user to reservation the driver and includes the required information to accept the order	
TC002	User select appropriate time	The system should display a screen that provide a different times to choose	
TC003	User Retrieve Reservation Information	This test case verifies that a user can successfully retrieve their reservation information	
TC004	User confirms booking	booking should be confirmed successfully	

Private Driver Services



SPRINT TWO



1. What has been completed since the last meeting?

- Log In Feature:
 - Implemented user login functionality, allowing users to authenticate with their email and password. Included backend authentication with bcrypt password comparison.
 - Created error handling for invalid email or password attempts.
- Lounge Reservation Feature:
 - Developed the backend API to allow users to search for and reserve airport lounges.
 - Designed the lounge reservation UI in Flutter, allowing users to choose lounge locations, input their details, and confirm their reservation.

2. What are you going to be working on next?

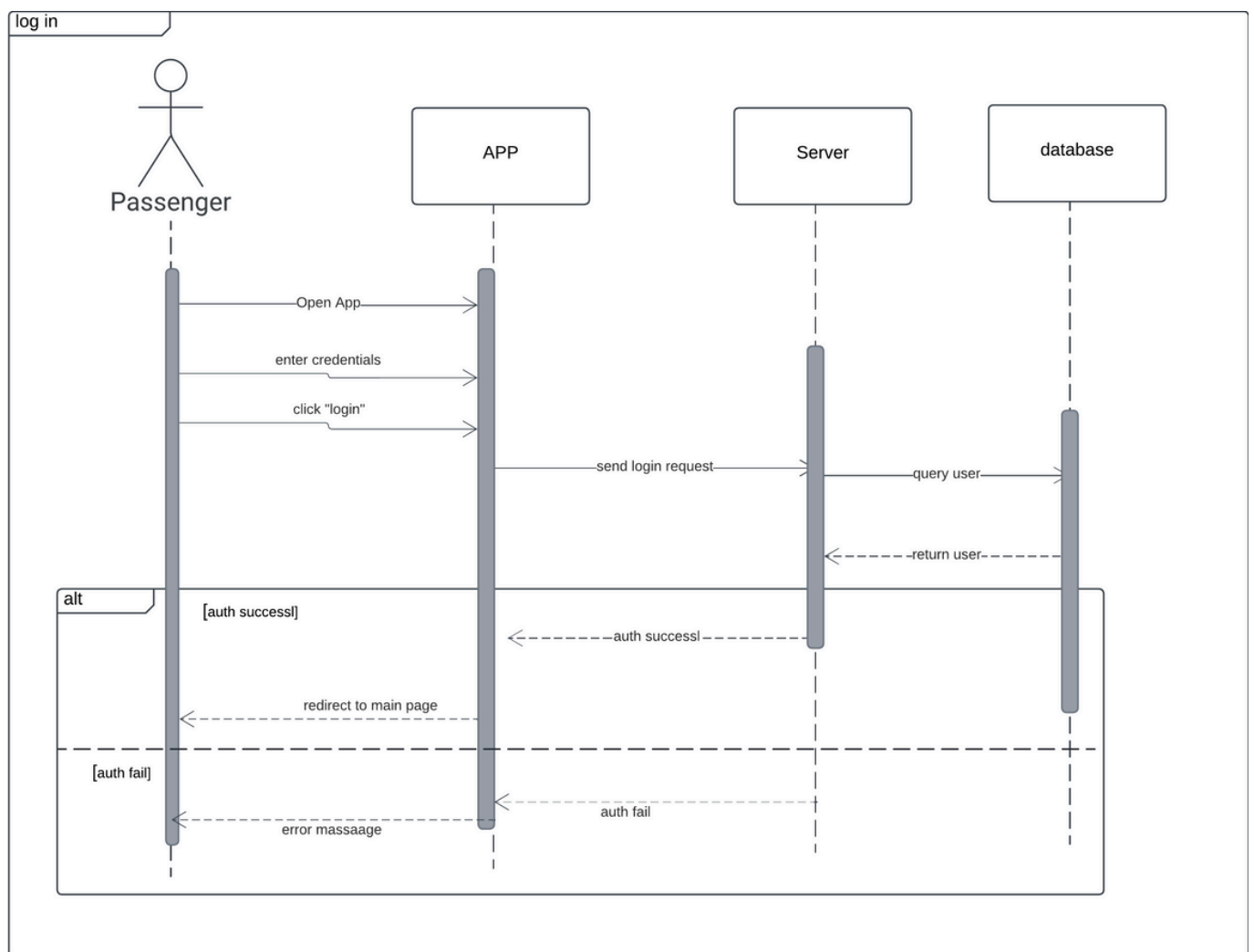
- Parking Reservation Feature: Start working on the parking reservation functionality, where users can book parking spots at the airport.
- Personal Companion Service: Begin the development of the Personal Companion Service feature, allowing users to request support for elderly or special needs travelers.

3. Do you have any issues/impediments?

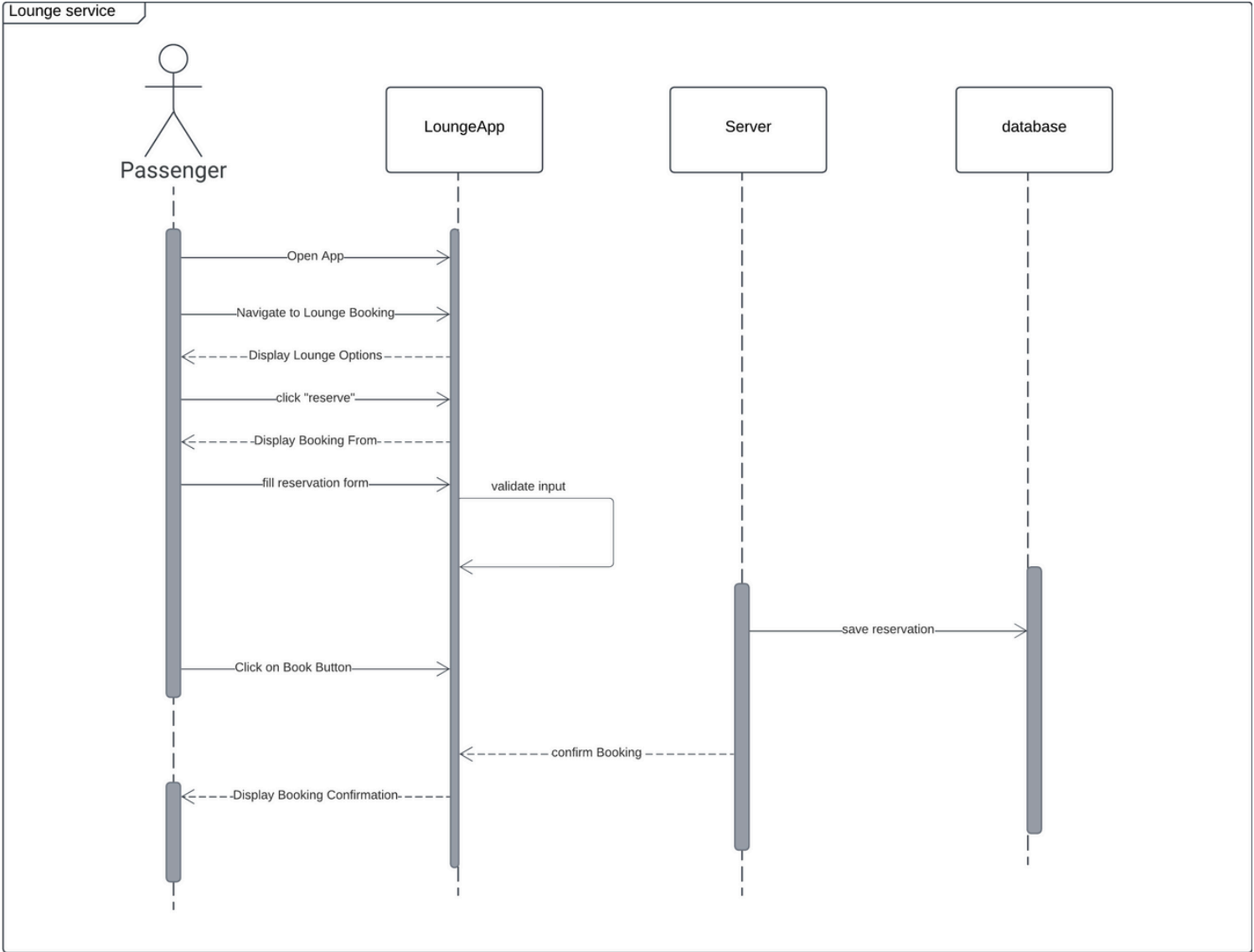
- API Response Issues: Occasional delays in fetching lounge reservation data from the backend; further optimizations may be required.

Story backlog:

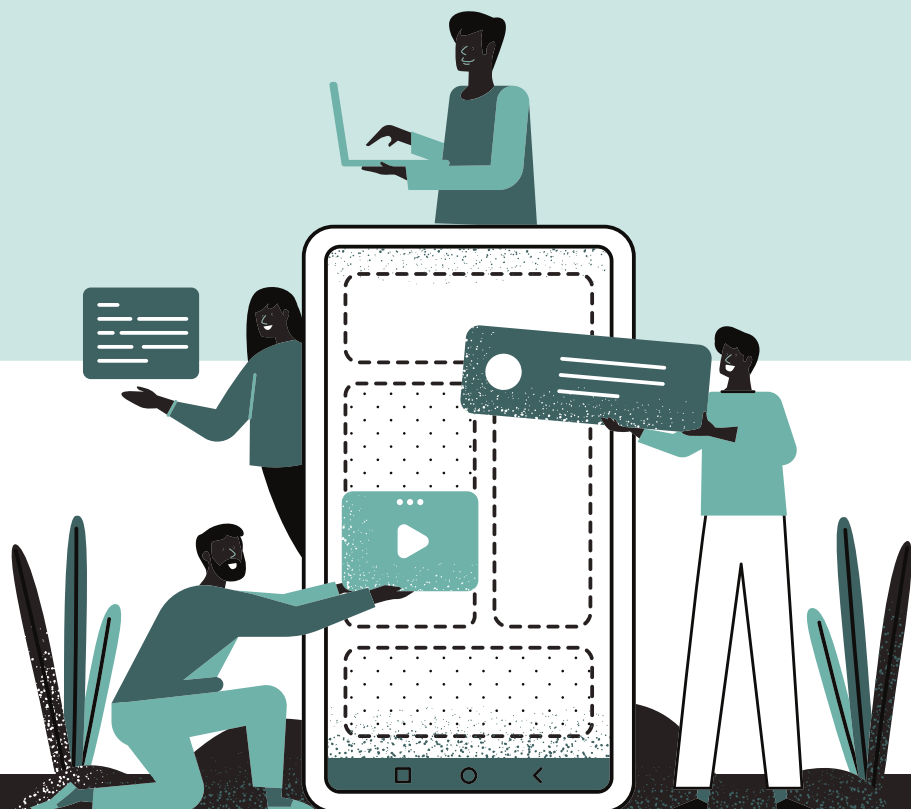
Story Name	Login
Story Sequence No	001
Story Short Description	Users must be able to log into the application using their username and password.
Story Long Description	The system allows users to enter their email and password to access the application. Upon successful login, users are directed to the main services page.



Story Name	Lounge Reservation
Story Sequence No	005
Story Short Description	Users can search for and reserve airport lounges.
Story Long Description	The system shall allow users to search for available airport lounges, input their personal information, select a desired lounge and departure time, and complete the booking process. The booking confirmation shall be displayed to the user, and the reservation details shall be stored in the user's profile for easy access.



SPRINT THREE



1. What has been completed since the last meeting?

- Parking Reservation Feature:
 - Developed the backend API for parking reservations, allowing users to select a parking spot, specify their arrival date/time, and reserve a parking space.
 - Implemented the parking reservation UI in Flutter, where users can select parking spots, enter reservation details, and confirm their booking.
- Personal Companion Service:
 - Created the backend API to support the Personal Companion Service, where users can request assistance for elderly, unaccompanied minors, or passengers with special needs.
 - Developed the UI in Flutter to allow users to request this service by entering personal details and selecting the type of assistance required

2. What are you going to be working on next?

- "Previous Reservations" Feature: Start developing the "Previous Reservations" feature, allowing users to view their past reservations for lounges, parking, and private drivers.
- Interface Enhancements: Focus on improving the overall user interface, making it more intuitive and visually appealing.

3. Do you have any issues/impediments?

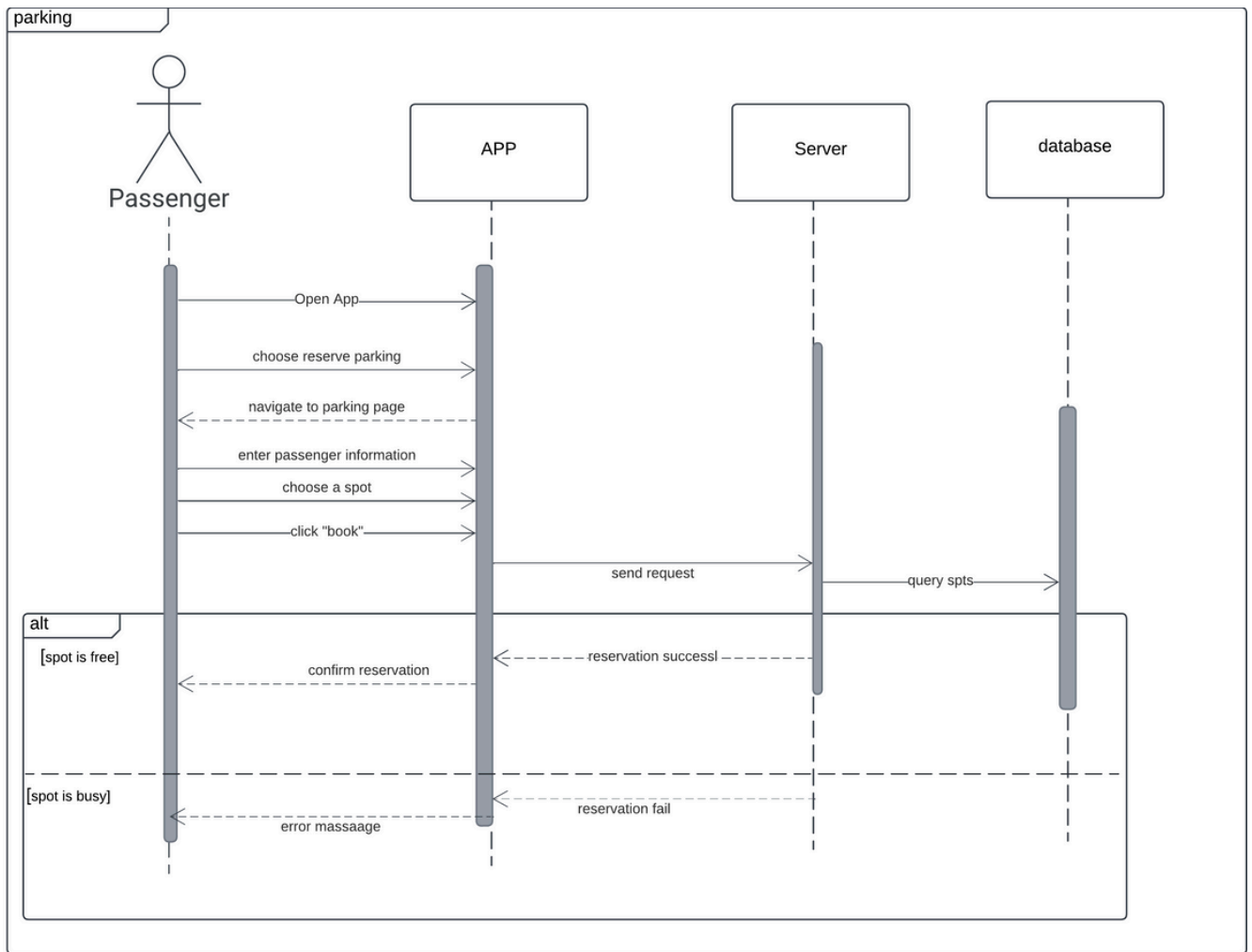
- Complex Data Validation: Data validation for the Personal Companion Service can be complex due to multiple input fields (e.g., type of traveler, date, etc.). Some edge cases need further refinement.
- Performance Issues: Performance bottlenecks with the backend queries, especially when fetching multiple reservation data for parking spots. Need to look into query optimization.

Story backlog:

Story Name	Parking Reservation
Story Sequence No	004
Story Short Description	Users can reserve airport parking.
Story Long Description	The system shall allow users to select and reserve parking spaces in advance. Users will be able to provide details such as name, phone number, email address, vehicle number, arrival date and time, reservation period, and select a parking spot from a visual grid. All reservations will be stored in the database.

Test Case Name: [Sprint 3 – Reserve parking]

Test Case NO.	Test Case Description	Expected Results	Outcome (Pass/Fail/Other(Comments))
TC001	Valid request	Created parking reservation	
TC002	Test reservation submission with missing name	The operation will not be completed until the user enter his name	
TC003	Test reservation submission with invalid Email	Error message will display "Email can not be empty"	
TC004	Test reservation with missing date	Error message "please select an arrival date"	

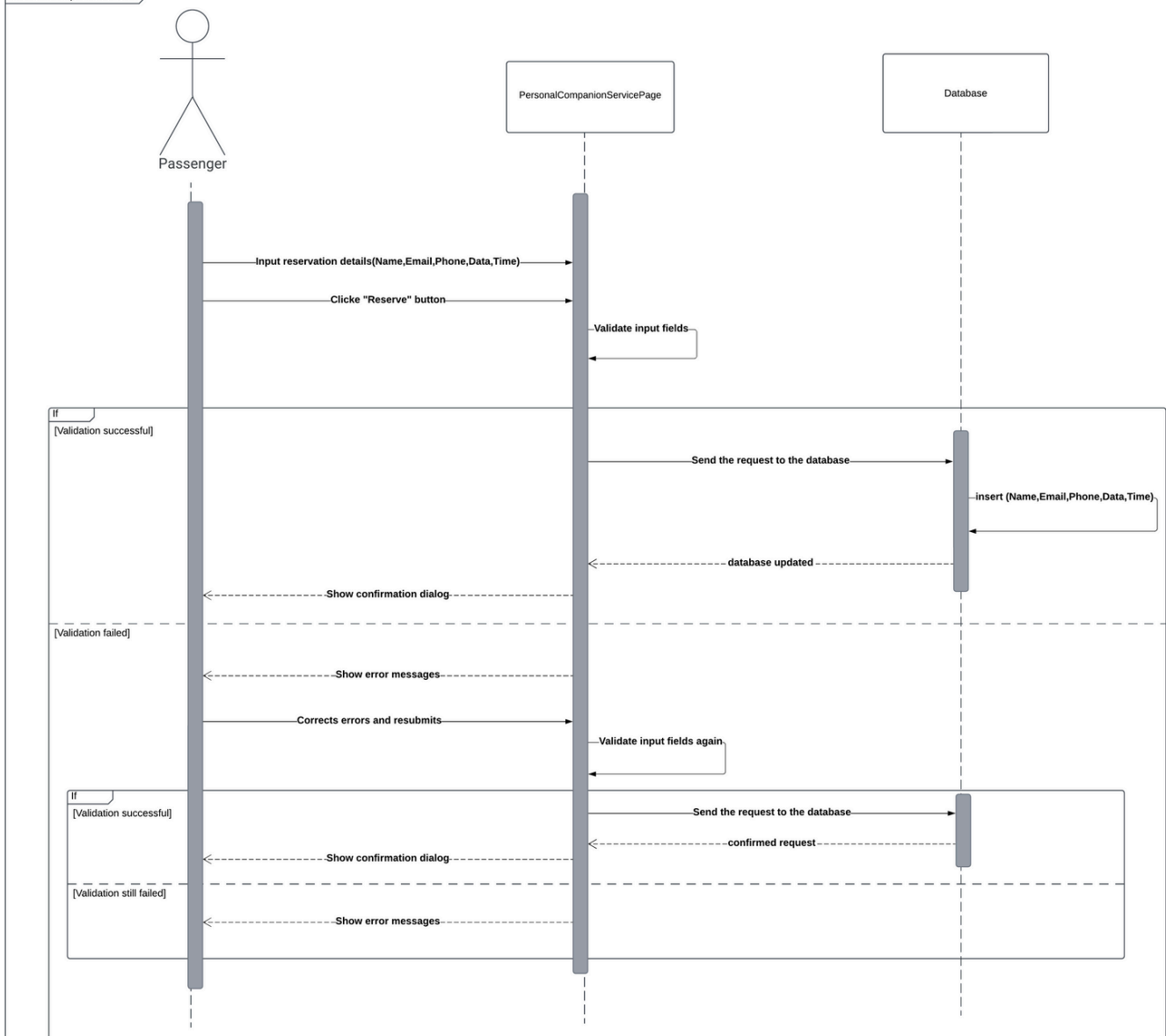


Story Name	Personal Companion Service
Story Sequence No	006
Story Short Description	Offer support services for travelers with special needs.
Story Long Description	The service shall assist elderly travelers, unaccompanied minors, and passengers with special needs to ensure their safety and support while navigating the airport. The application shall allow users to request this service by providing their details, such as name, email, phone number, and arrival date/time. Users will also be able to select the type of traveler (e.g., child or special need) through a toggle button.

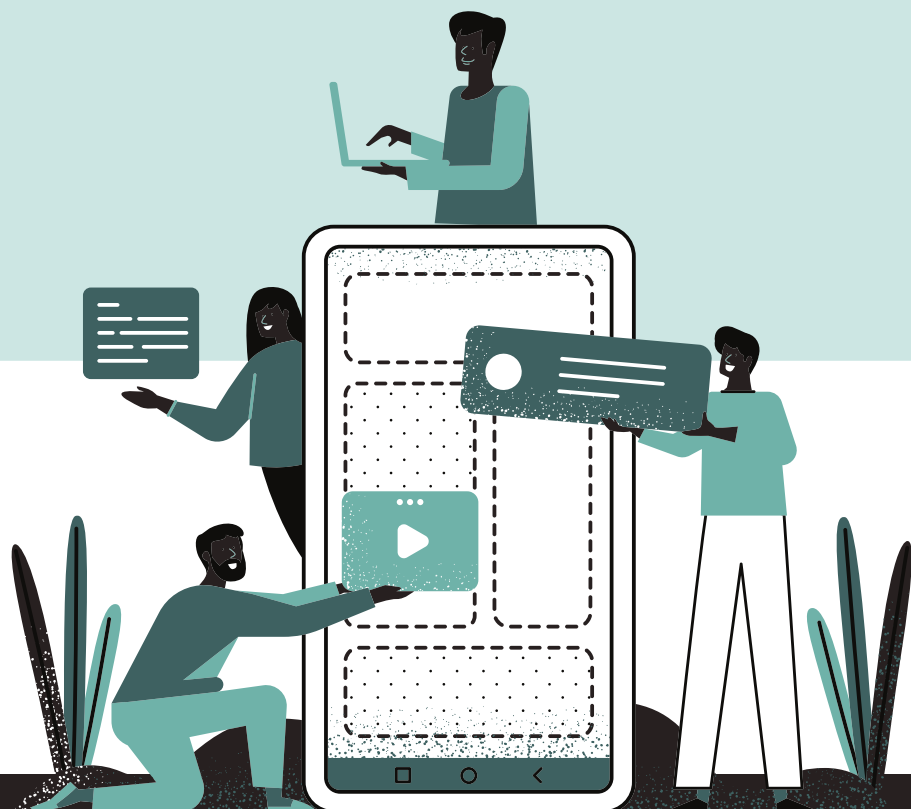
Test Case Name: [Sprint 3 – Personal Companion Service]

Test Case NO.	Test Case Description	Expected Results	Outcome (Pass/Fail/Other(Comments))
TC001	Test successful reservation submission.	Confirmation dialog shows Reservation ID.	
TC002	Test reservation submission with missing name.	Error message: "Name cannot be empty".	
TC003	Test reservation submission with invalid email.	Error message: "Email cannot be empty".	
TC004	Test reservation submission with missing phone.	Error message: "Phone number cannot be empty".	
TC005	Test reservation submission with missing date.	Error message: "Please select an arrival date".	
TC006	Test reservation submission with missing time.	Error message: "Please select an arrival time".	

Personal Companion Service



FINAL SPRINT



1. What has been completed since the last meeting?

- "Previous Reservations" Feature:
 - Completed the "Previous Reservations" feature, which allows users to view their past reservations across lounges, parking, and private drivers. The backend API was updated to fetch past reservation data based on the user's email.
 - Integrated the feature in the app, displaying the data in a user-friendly dialog with the option to view past bookings.
- Interface Enhancements:
 - Refined the UI for reservation dialogs, including the "Previous Reservations" dialog and the user info dialog.
 - Enhanced the design for better user experience, improved button styles, and ensured mobile responsiveness.

2. What are you going to be working on next?

- Final Testing & Bug Fixes: Conduct final testing for all features to ensure everything works as expected and fix any remaining bugs or performance issues.
- Deployment: Prepare the app for deployment by addressing any remaining technical debt, completing documentation, and finalizing the build.

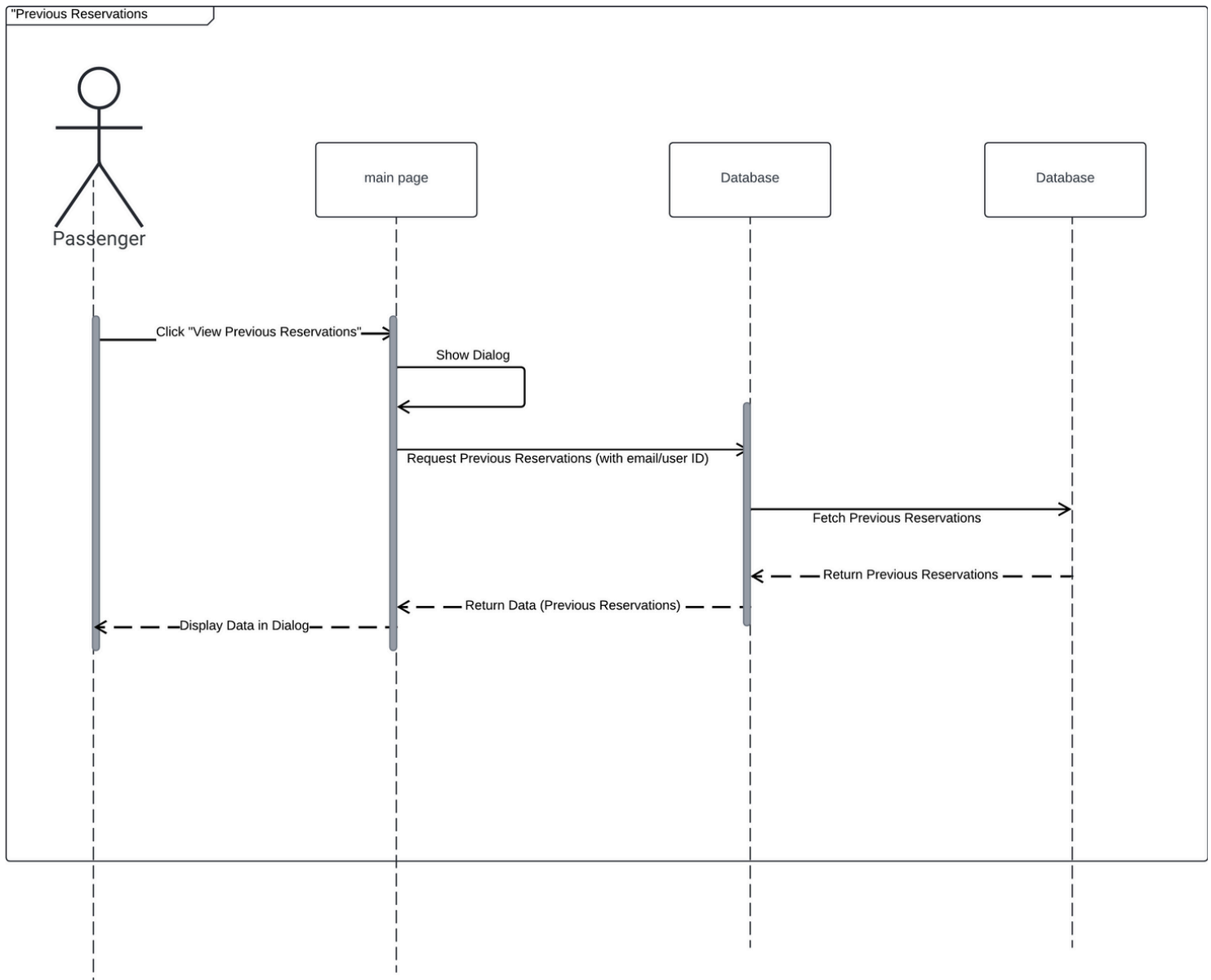
3. Do you have any issues/impediments?

- Database Query Optimization: The "Previous Reservations" feature is still facing performance issues when retrieving large sets of data. Pagination or further query optimization will be required for better performance.
- Final UI Tweaks: Need some final tweaks to ensure consistency across the app, particularly for smaller devices.

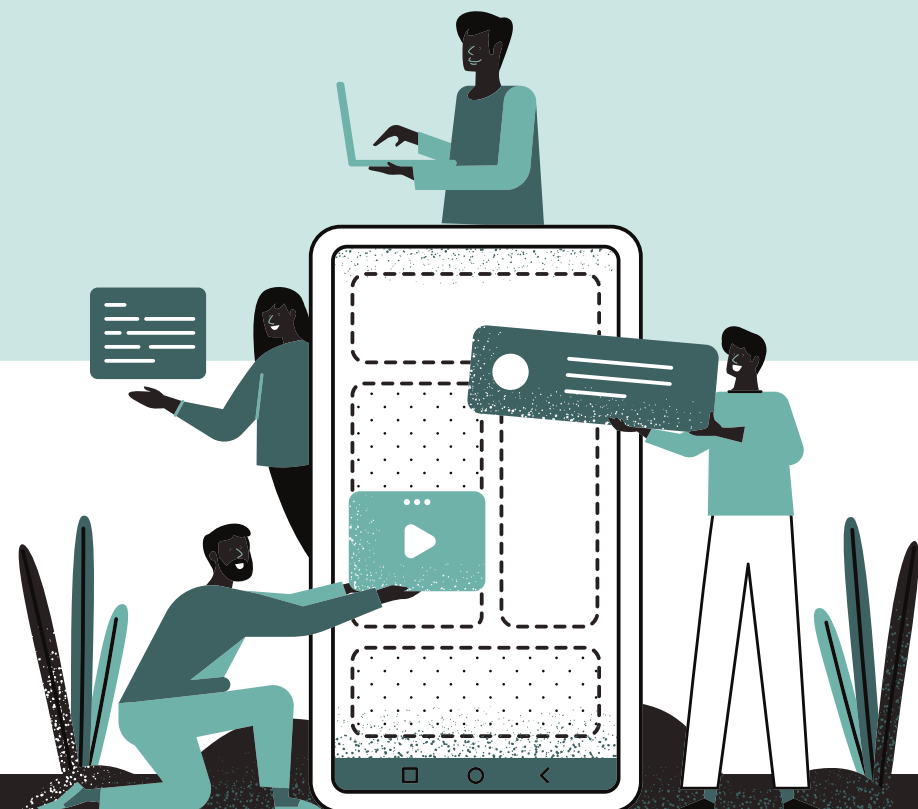
Previous Reservations Feature

Feature Overview

The Previous Reservations feature has been added to the application to allow users to easily view and manage their past bookings. This feature provides users with a quick way to check their previous reservations across various services, such as lounges, parking spots, and private drivers, all from within the .app

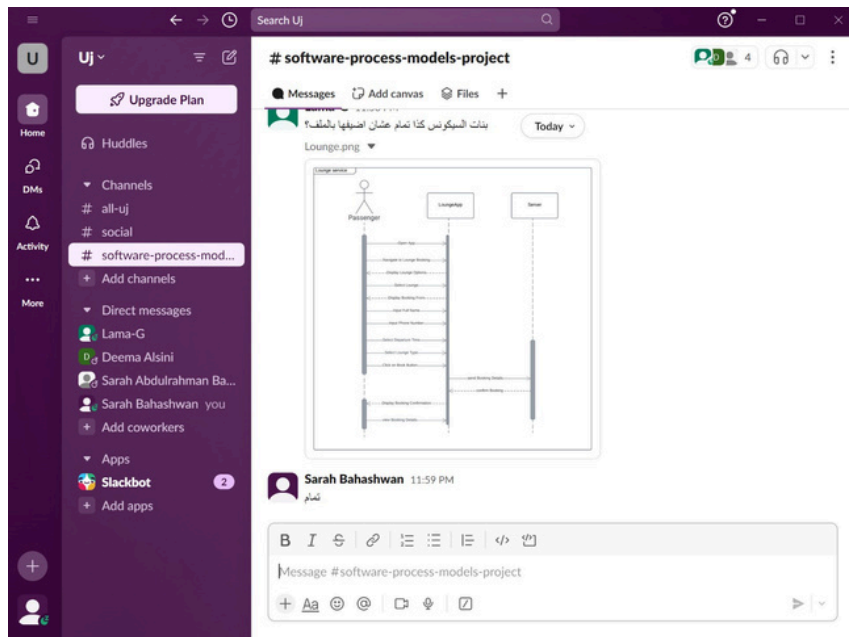


WORK FLOW



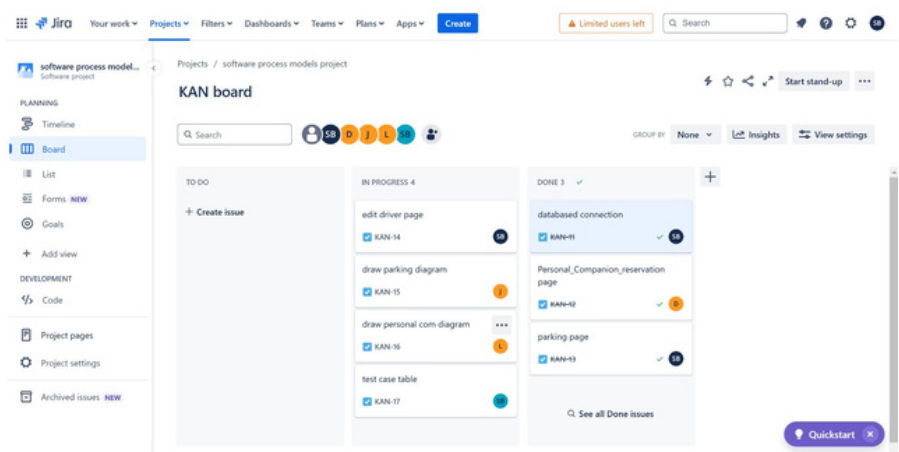
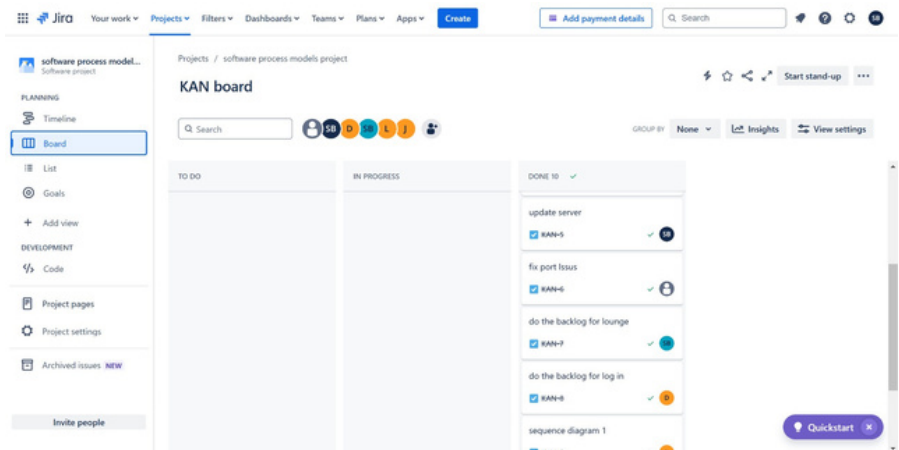
slack

We used slack for team communication as it helped us discuss within team in a efficient way



jira

jira was the tool we used for tracking workflow and managing tasks, helping us watch progress, keep stories prioritized, and align with project objectives.



Conclusion

The airport service application successfully addresses key pain points for passengers by integrating essential features like personal companion services, lounge reservations, parking space booking, and private driver arrangements. These services ensure a seamless and stress-free travel experience for users, prioritizing convenience, safety, and efficiency. By centralizing these offerings within a single platform, the application significantly enhances the overall travel journey for both occasional travelers and frequent flyers, making air travel more enjoyable and efficient.

Future Plans

To further elevate the functionality and user experience of the platform, we aim to introduce the following features:

1. **3D Airport Navigator:**

- A detailed 3D map of the airport to help passengers easily locate gates, lounges, restrooms, and other essential facilities.
- Turn-by-turn navigation to guide users through the airport, reducing confusion and delays.

2. **Real-Time Flight and Baggage Tracker:**

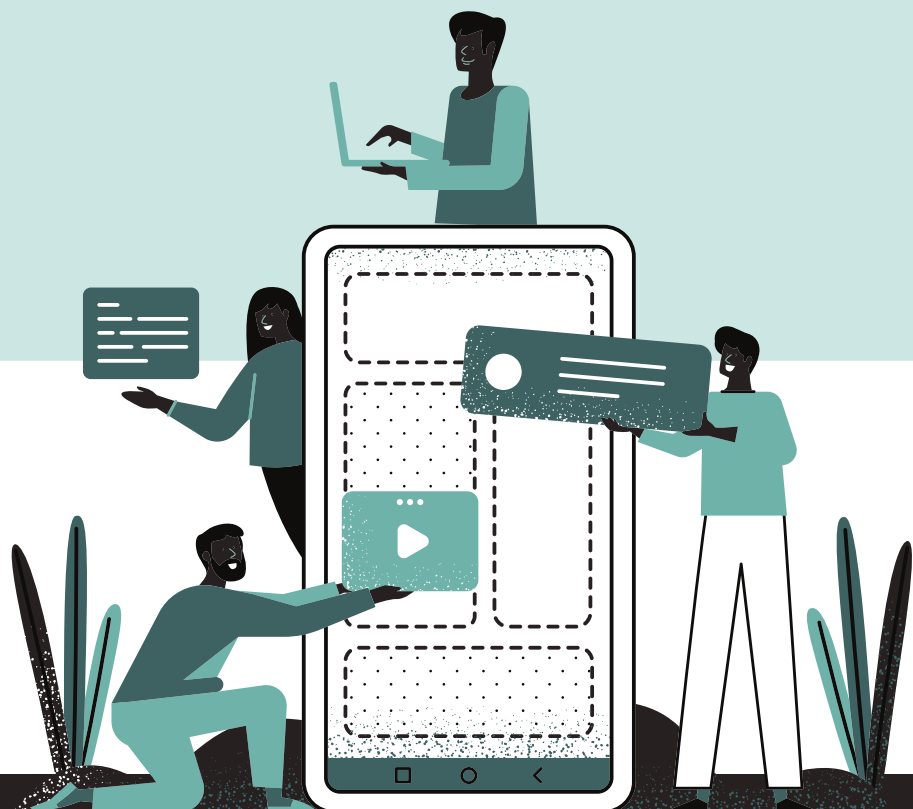
- Integration with airline systems to provide live updates on flight schedules, delays, and gate changes.
- A baggage tracker to allow passengers to monitor their luggage status from check-in to collection, ensuring peace of mind.

3. **Time and Currency Converter:**

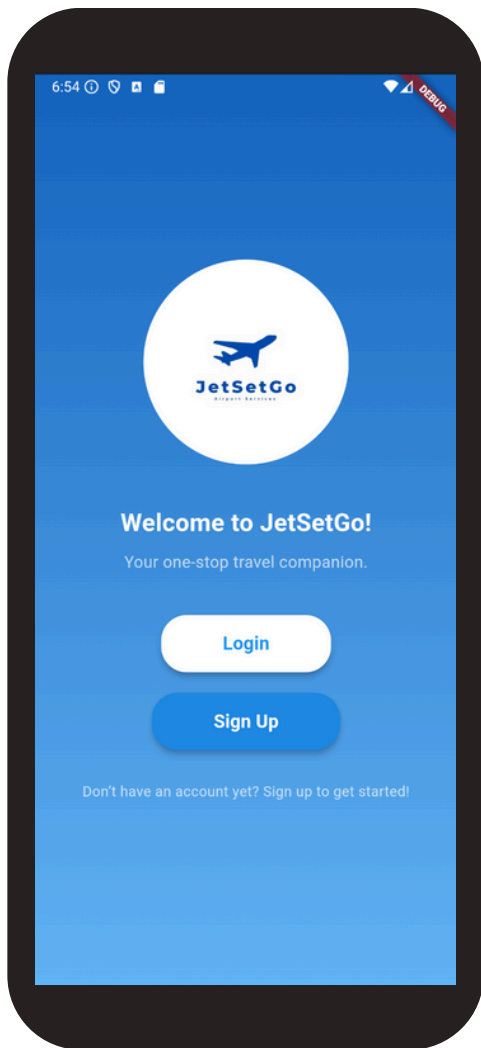
- A built-in tool to help passengers quickly adjust to different time zones and calculate currency exchange rates.
- Useful for international travelers managing schedules and budgets while on the move.

These planned features aim to transform the application into a comprehensive digital travel assistant, setting a new standard for airport service platforms. By continuously innovating, we seek to cater to a broader audience, improve operational efficiency, and stay ahead in the competitive travel technology landscape.

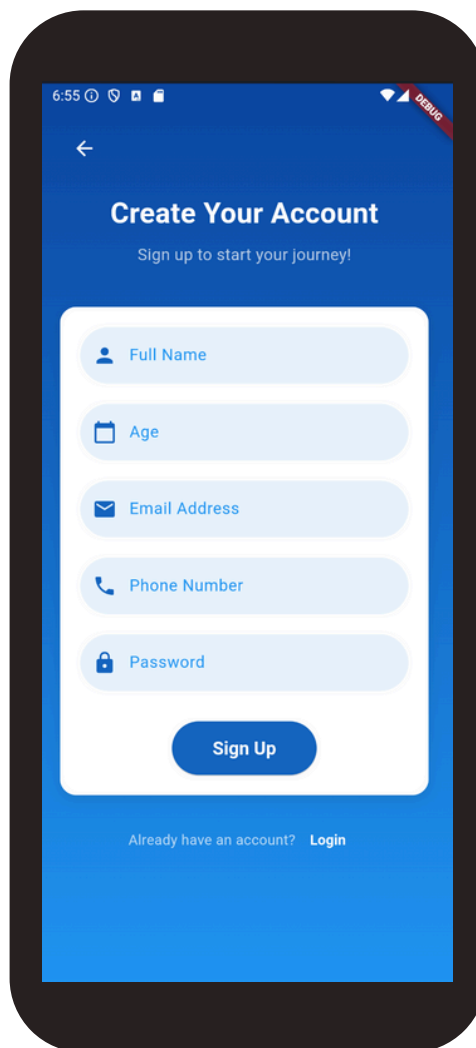
INTERFACES



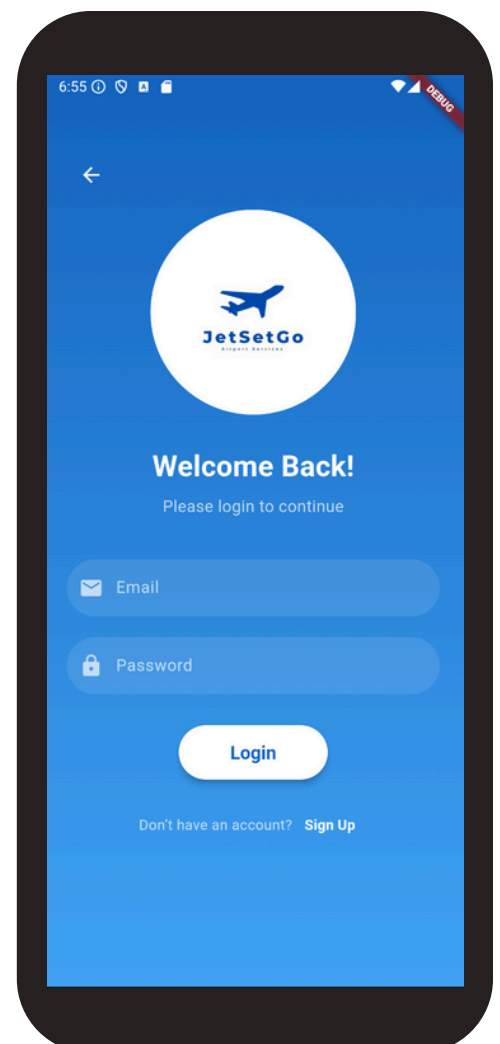
welcome page



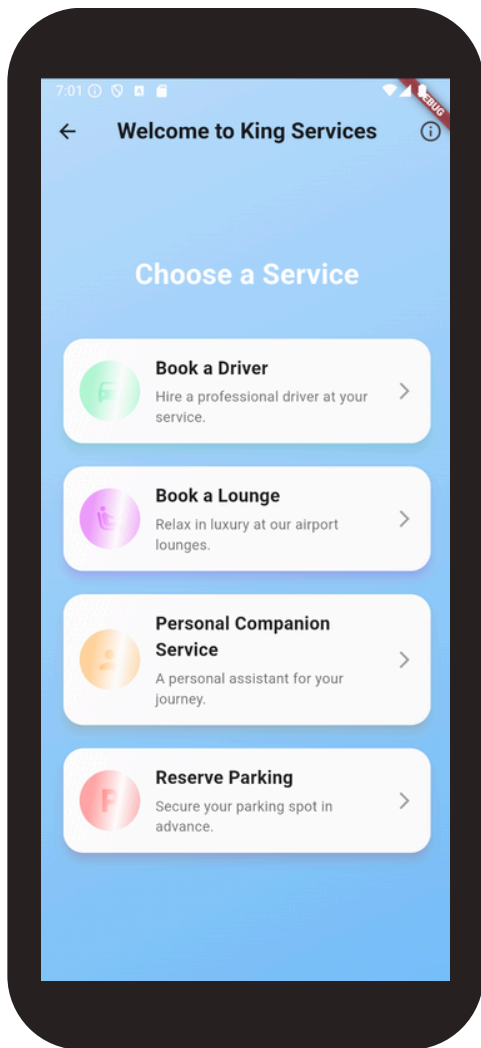
sign up page



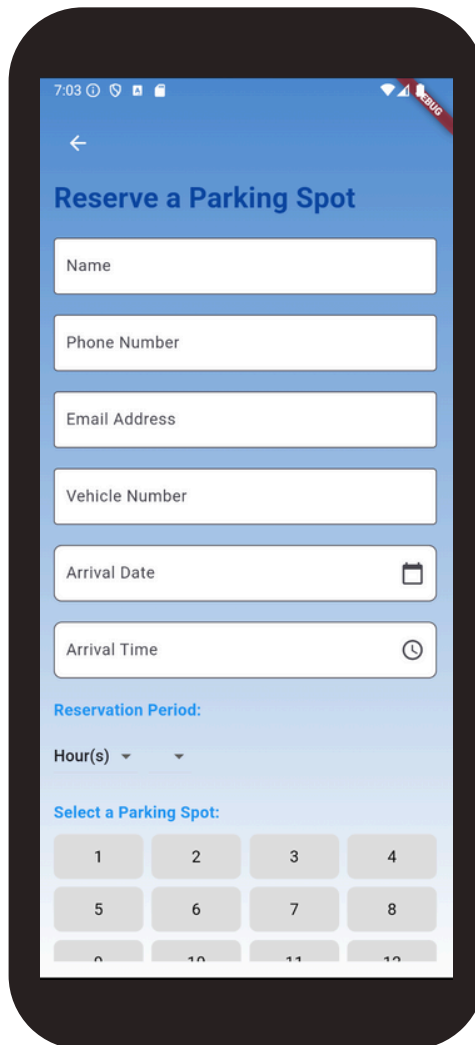
log in page



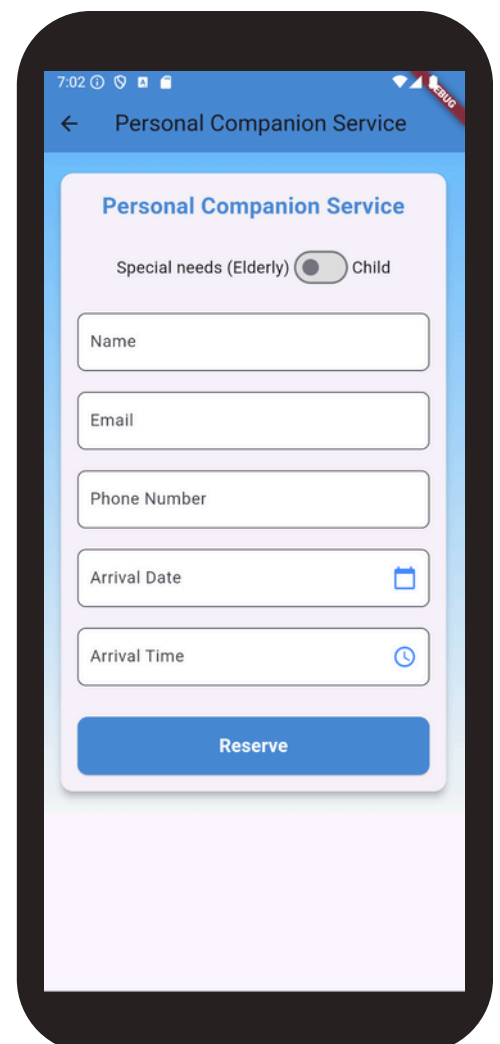
main page



parking page



personal companion page



[illegible]

A screenshot of a mobile application interface for King Abdulaziz Airport Lounges. The top status bar shows the time 7:01 and various icons. The app's header is blue with a back arrow and the title "King Airport Lounge". Below the header is a large image of the airport terminal at night. The main heading is "King Abdulaziz Airport Lounges" in bold black text, followed by the subtitle "Select a lounge to access premium services." in a lighter font. There are three lounge options listed in white rounded rectangular cards: 1. "AL Fursan Lounge" with a blue circular icon containing a white airplane and the text "AL FURSAN", description "Premium comfort and services.", and a right arrow. 2. "PLAZA Premium Lounge" with a blue circular icon containing a white coffee cup and the text "PLAZA", description "Relax and rejuvenate in style.", and a right arrow. 3. "Welcome Lounge" with a blue circular icon containing a white bed and the text "WELCOME", description "Exceptional hospitality awaits.", and a right arrow.

A screenshot of a mobile application interface for Visa Lounges. The top status bar shows the time 7:01 and various icons. The app's header is blue with a back arrow and the text "Lounge 1". A red diagonal banner in the top right corner says "New". The main content area has a light blue background. In the center, there is a white rounded rectangle containing the Plaza Premium Lounge logo (a circular emblem with a central sun-like design) and the text "PLAZA PREMIUM LOUNGE" in a bold, dark purple font. Below this, a paragraph of text states: "Some lounges may require eligibility verification at the entrance. Please ensure you have the required credentials." Underneath the paragraph, the heading "Accepted VISA Types:" is followed by a list of four Visa card types: "VISA Classic", "VISA Gold", "VISA Platinum", and "VISA Signature", each preceded by a hyphen. At the bottom of the screen, there is a blue button with the text "Go to Lounge Form".

lounge 3 page

7:02

← Lounge Booking

Book Your Lounge

Secure your lounge reservation with the form below.

Full Name

Email

Phone Number

Departure Date

Select Lounge

Book Now

Previous Reservations dialog

7:03

← Welcome to King Services

Choose a Service

Book a Driver
Hire a professional driver at your service.

Reserve Parking
Secure your parking spot in advance.

More Options

Previous Reservations

Cancel



student name:

sarah bahashwan

deema alsini

joud alahmadi

lama alghamdi

sarah bajarsh